

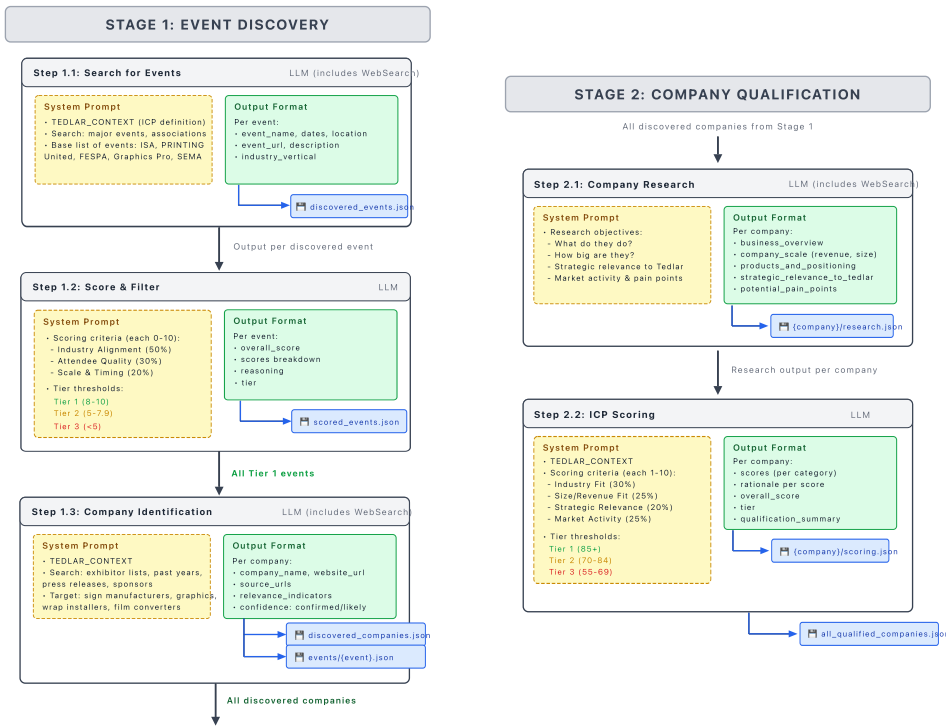
# DuPont Tedlar AI Lead Generation System

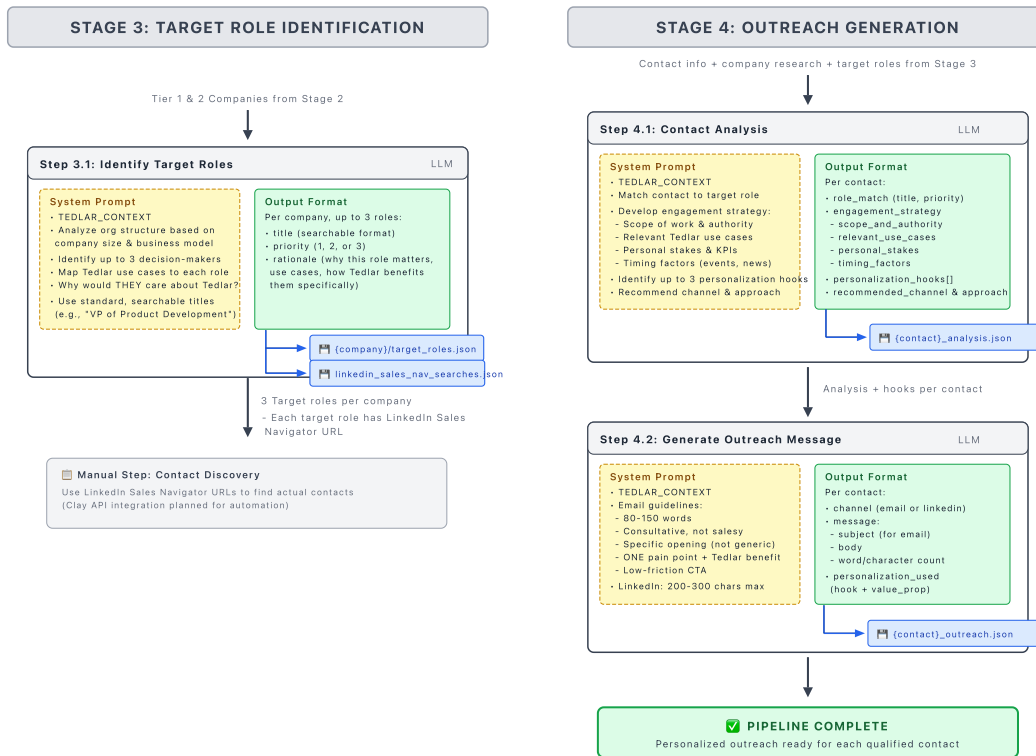
## 1. Solution Architecture

A **four-stage AI GTM workflow** that progressively identifies, scores and filters leads:



## 2. Data Processing Pipeline





### 3. AI Workflow Orchestration

Pipeline is orchestrated by `main.py` that relies on event triggers with:

- Priority order:** Later stages processed first such that companies closer to completion are processed through the pipeline before starting new ones.
- Automatic triggers:** Callbacks connect stages (when Stage 2 completes, Stage 3 starts automatically for qualified companies).
- Rate limiting:** Shared limiter tracks API tokens (30k/minute) across all stages.
- Resume capability:** Each stage produces JSON artifacts, enabling resume from interruption and preventing duplicate reruns.
- Tiered qualification:** Scoring thresholds before proceeding to next stage ensures resources focus on high-potential leads. Low-scoring companies filtered early.

## 4. Implementation Results

### 4.1 Data Outputs

Each stage produces JSON artifacts stored in `data/`:

```
data/
├── events/
│   ├── discovered_events.json    # 22 raw events from Stage 1.1
│   ├── scored_events.json       # Tiered events from Stage 1.2
│   ├── discovered_companies.json # 37 companies from Stage 1.3
│   └── companies/
│       └── {EventName}.json      # Companies per event
└── companies/
```

```
|
| | tier_1_high_priority.json    # Summary of top companies
| | all_qualified_companies.json
| | {CompanyName}/
| | | research.json            # Stage 2.1 deep research
| | | scoring.json             # Stage 2.2 ICP scores
| | | target_roles.json        # Stage 3 decision-makers
| | | linkedin_searches.json   # Sales Nav search URLs
| | contacts/
| | | role_identification_results.json
| | | linkedin_sales_nav_searches.json
| | | {CompanyName}/
| | | | {ContactName}_analysis.json    # Stage 4.1 hooks/angles
| | | | {ContactName}_outreach.json    # Stage 4.2 email draft
```

## 5. Limitations & Future Work

### Current Limitations:

- Contact discovery requires manual LinkedIn searches (Clay API integration planned but not implemented yet)
- Rate limits require ~60s min. delays between API calls

### Potential Enhancements:

- Integrate Clay API for automated contact enrichment
- Add email verification step before outreach
- CRM integration for direct lead export